

MA-202

Challenging Problems: Jensen's Inequality & Standard Distributions

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Part A: Jensen's Inequality & Theoretical Bounds

Q1. (Jensen's Inequality for Beta) Let $X \sim \text{Beta}(\alpha, \beta)$ with $\alpha, \beta > 0$.

- (a) Using Jensen's inequality, prove that $E[\ln X] < \ln\left(\frac{\alpha}{\alpha+\beta}\right)$.
- (b) Verify if the inequality holds for the limiting case where the variance approaches 0.

Q2. (Lyapunov's Inequality) Let X be a random variable such that $X > 0$ almost surely. Using Jensen's inequality (or otherwise), prove that for any $s > r > 0$:

$$(E[X^r])^{1/r} \leq (E[X^s])^{1/s}$$

Hint: Apply Jensen's inequality to the convex function $g(t) = t^{s/r}$ with the random variable $Y = X^r$.

Q3. (The Jensen Gap) Let $X \sim U(1, 2)$.

- (a) Calculate the exact value of $E[1/X]$ and $1/E[X]$.
- (b) Verify that $E[1/X] > 1/E[X]$ and calculate the magnitude of the "Jensen Gap" $\Delta = E[1/X] - 1/E[X]$.

Part B: Distribution-Specific Challenges

Q4. (Moments of Reciprocal Beta) Let $X \sim \text{Beta}(\alpha, \beta)$.

- (a) Prove that for $k \in \mathbb{Z}^+$, the k -th moment of $1/X$ exists if and only if $\alpha > k$.
- (b) Specifically, show that if $\alpha > 2$:

$$\text{Var}\left(\frac{1}{X}\right) = \frac{\beta(\alpha + \beta - 1)}{(\alpha - 1)^2(\alpha - 2)}$$

Q5. (Transformation of Variables) Let $X \sim U(0, 1)$.

- (a) Find the probability density function (PDF) of the random variable $Y = -2 \ln X$.
- (b) Identify the distribution of Y and use this relationship to find $P(Y > 4)$ without integrating the PDF of Y .

Q6. (Hierarchical Models) Let $X \sim U(0, 1)$. Given $X = x$, let Y be uniformly distributed on the interval $[0, x]$ (i.e., $Y|X \sim U(0, x)$).

- (a) Find the marginal PDF of Y , $f_Y(y)$.
- (b) Calculate the unconditional expectation $E[Y]$ and variance $\text{Var}(Y)$.

Q7. (Normal Parameters from Quantiles) Let $X \sim N(\mu, \sigma^2)$. Given:

$$P(X < 35) = 0.07 \quad \text{and} \quad P(X < 63) = 0.89$$

Find the exact values of μ and σ . (Use $\Phi^{-1}(0.07) \approx -1.476$ and $\Phi^{-1}(0.89) \approx 1.227$).

Q8. (Tail Probabilities) Let $X \sim N(11, 2.25)$ (where $\sigma = 1.5$).

- (a) Find x_0 such that $P(X > x_0) = 0.09$.
- (b) Compare this x_0 with the bound provided by Chebyshev's inequality for the same probability. i.e., Find k such that $P(|X - \mu| \geq k\sigma) \leq 0.09$ implies a certain deviation.

Q9. (Variance Matching) Let $X \sim U(-1, 3)$ and $Y \sim \text{Exp}(\lambda)$.

- (a) Find the value of λ such that $\text{Var}(X) = \text{Var}(Y)$.
- (b) For this λ , calculate $P(Y > E[X])$.

Q10. (Expected Distance) Two points X and Y are chosen independently and uniformly from the interval $[0, 1]$. Find the expected distance between them, $E[|X - Y|]$. (*Hint: This requires double integration over the unit square*).

Solutions & Derivations

Solution 1: Jensen's on Beta

(a) The function $g(x) = \ln x$ is strictly concave ($\frac{d^2}{dx^2} \ln x = -1/x^2 < 0$ for $x > 0$). Jensen's inequality for a concave function states $E[g(X)] \leq g(E[X])$. Since $X \sim \text{Beta}(\alpha, \beta)$ is non-degenerate (assuming variance > 0), the inequality is strict:

$$E[\ln X] < \ln(E[X])$$

We know $E[X] = \frac{\alpha}{\alpha+\beta}$. Therefore, $E[\ln X] < \ln\left(\frac{\alpha}{\alpha+\beta}\right)$.

(b) The inequality becomes an equality only if X is a constant (degenerate). This happens when the variance is 0, which requires $\alpha \rightarrow \infty$ or $\beta \rightarrow \infty$ such that the distribution converges to a point mass.

Solution 2: Lyapunov's Inequality

Let $Y = X^r$. Since $X > 0$, $Y > 0$. We want to relate $E[X^s]$ to $E[X^r]$. Note that $X^s = (X^r)^{s/r} = Y^{s/r}$. Let $\phi(y) = y^{s/r}$. Since $s > r$, the exponent $p = s/r > 1$. The function $\phi(y) = y^p$ is convex for $y > 0$ and $p > 1$ (second derivative $p(p-1)y^{p-2} > 0$). By Jensen's Inequality: $E[\phi(Y)] \geq \phi(E[Y])$. Substituting back:

$$E[X^s] \geq (E[X^r])^{s/r}$$

Raising both sides to the power $1/s$ (which is positive, preserving inequality):

$$(E[X^s])^{1/s} \geq (E[X^r])^{1/r}$$

Solution 3: The Jensen Gap

(a) $X \sim U(1, 2)$, so $f(x) = 1$ for $1 \leq x \leq 2$. $E[X] = \frac{1+2}{2} = 1.5$. Thus, $1/E[X] = 2/3 \approx 0.6667$. $E[1/X] = \int_1^2 \frac{1}{x} \cdot 1 dx = [\ln x]_1^2 = \ln 2 \approx 0.6931$. (b) Since $g(x) = 1/x$ is convex, $E[1/X] > 1/E[X]$. Gap $\Delta = \ln 2 - \frac{2}{3} \approx 0.6931 - 0.6667 = 0.0264$.

Solution 4: Variance of Reciprocal Beta

(a) $E[1/X^k] = \frac{1}{B(\alpha, \beta)} \int_0^1 x^{-k} x^{\alpha-1} (1-x)^{\beta-1} dx = \frac{1}{B(\alpha, \beta)} \int_0^1 x^{\alpha-k-1} (1-x)^{\beta-1} dx$. This integral converges to $B(\alpha-k, \beta)$ only if the exponent of x is > -1 , i.e., $\alpha-k-1 > -1 \implies \alpha > k$.

(b) We need $E[1/X]$ and $E[1/X^2]$. Using the property $\Gamma(z+1) = z\Gamma(z)$:

$$\begin{aligned} E\left[\frac{1}{X}\right] &= \frac{\alpha + \beta - 1}{\alpha - 1} \\ E\left[\frac{1}{X^2}\right] &= \frac{B(\alpha-2, \beta)}{B(\alpha, \beta)} = \frac{(\alpha + \beta - 1)(\alpha + \beta - 2)}{(\alpha - 1)(\alpha - 2)} \\ \text{Var} &= E\left[\frac{1}{X^2}\right] - \left(E\left[\frac{1}{X}\right]\right)^2 \\ \text{Var} &= \frac{(\alpha + \beta - 1)(\alpha + \beta - 2)}{(\alpha - 1)(\alpha - 2)} - \frac{(\alpha + \beta - 1)^2}{(\alpha - 1)^2} \end{aligned}$$

Factor out $\frac{\alpha+\beta-1}{\alpha-1}$:

$$= \frac{\alpha + \beta - 1}{\alpha - 1} \left[\frac{\alpha + \beta - 2}{\alpha - 2} - \frac{\alpha + \beta - 1}{\alpha - 1} \right]$$

Simplifying the bracket term yields $\frac{\beta}{(\alpha-1)(\alpha-2)}$. Result: $\frac{\beta(\alpha+\beta-1)}{(\alpha-1)^2(\alpha-2)}$.

Solution 7: Normal Parameters

We have the system: 1) $\frac{35-\mu}{\sigma} = Z_{0.07} \approx -1.476$ 2) $\frac{63-\mu}{\sigma} = Z_{0.89} \approx 1.227$ Subtract (1) from (2): $\frac{28}{\sigma} = 1.227 - (-1.476) = 2.703 \implies \sigma \approx 10.36$. Substitute σ into (2): $63 - \mu = 1.227(10.36) \approx 12.71 \implies \mu \approx 50.29$.

Solution 9: Variance Matching

(a) $X \sim U(-1, 3)$. Range = 4. $\text{Var}(X) = \frac{(b-a)^2}{12} = \frac{16}{12} = \frac{4}{3}$. $Y \sim \text{Exp}(\lambda)$. $\text{Var}(Y) = \frac{1}{\lambda^2}$. Set $\frac{1}{\lambda^2} = \frac{4}{3} \implies \lambda^2 = \frac{3}{4} \implies \lambda = \frac{\sqrt{3}}{2} \approx 0.866$. (b) $E[X] = \frac{-1+3}{2} = 1$. We need $P(Y > 1)$ for $Y \sim \text{Exp}(\sqrt{3}/2)$. $P(Y > k) = e^{-\lambda k}$. $P(Y > 1) = e^{-\sqrt{3}/2} \approx e^{-0.866} \approx 0.42$.